Fashion Sales & Customer Market Intelligence

A Data Science Project

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Tools: Python, Pandas, Matplotlib & Seaborn And Jupyter Notebook

INTRODUCTION

This project uses 5,000 simulated fashion sales records from 2024 to 2025.

The aim is to understand sales trends, customer behavior, and product performance to help the company make better business decisions.

DATASET OVERVIEW

- The dataset contains 5,000 sales records.
- It has 15 original columns.
- The data covers January 2024 to December 2025.
- It includes products, customers, prices, regions, and sales channels.
- The dataset was created for educational purposes.

DATA CLEANING

- Checked the dataset for missing values.
- Found 28 missing values.
- Replaced missing values with "Unknown".
- Converted order dates into the correct format.
- Checked for duplicate records.
- Saved the cleaned dataset.

FEATURE ENGINEERING

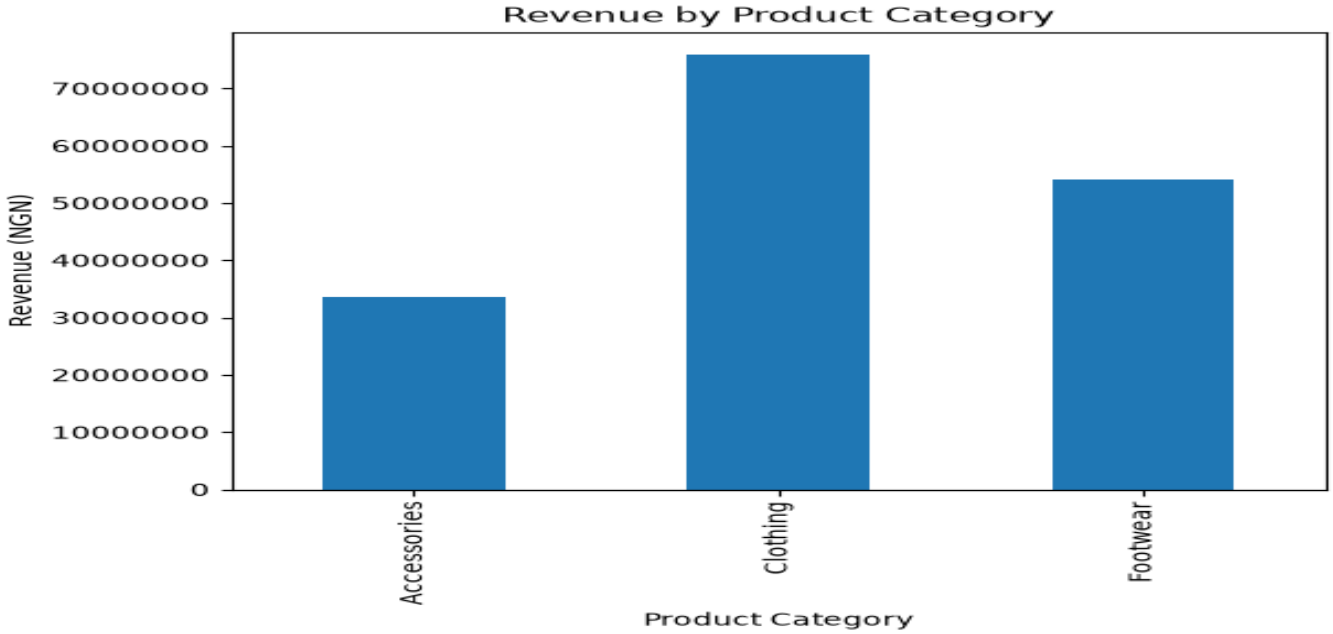
Original columns
15
New columns
7
Total after feature engineering: 22 columns, if all seven new columns are present.
New columns created

- Gross Sales
- Discount Amount
- Net Sales
- Month
- Month Name
- Quarter
- Season

- Average Order Value: ₦41,179.10

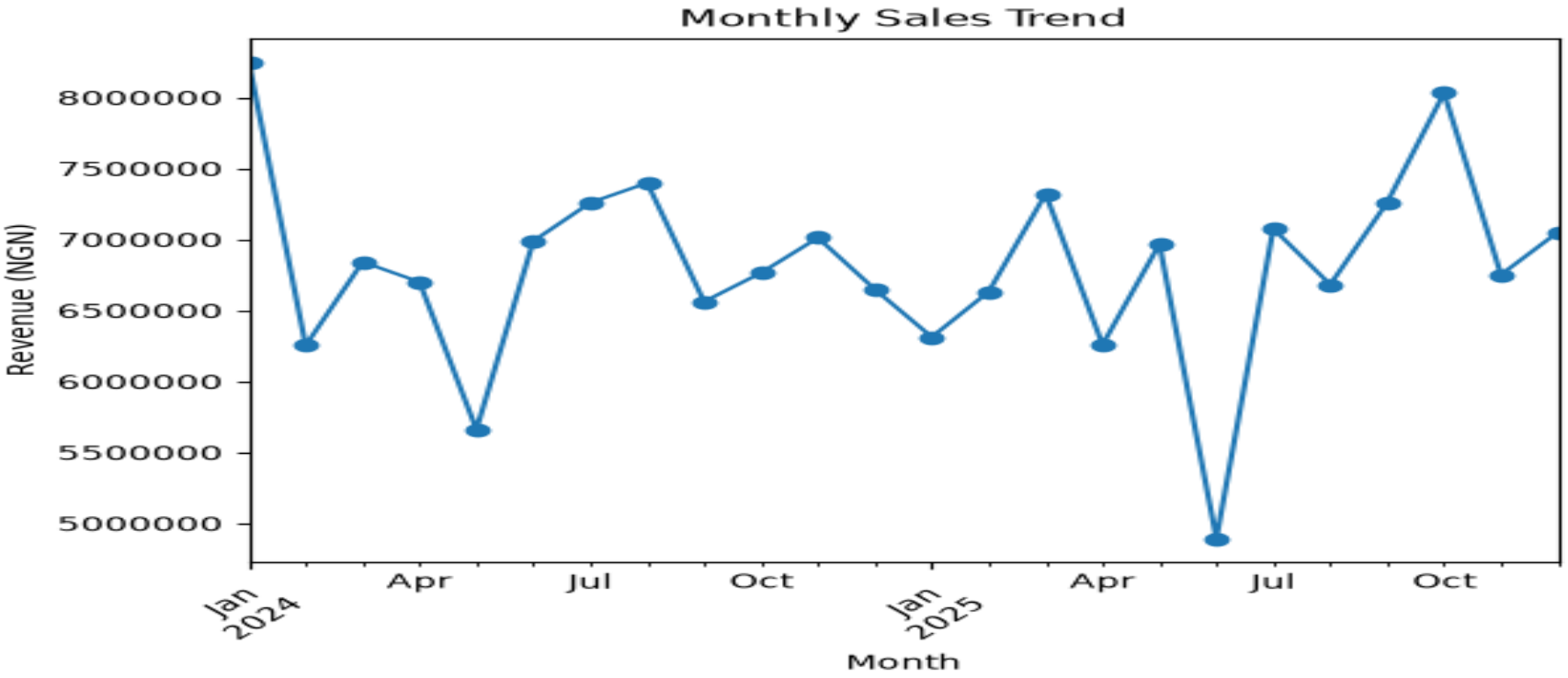
Sales Analysis

- Clothing generated the highest revenue at ₦75.99 million, while Accessories generated the lowest at ₦33.57 million. The company should maintain clothing stock and promote the other categories to increase sales.



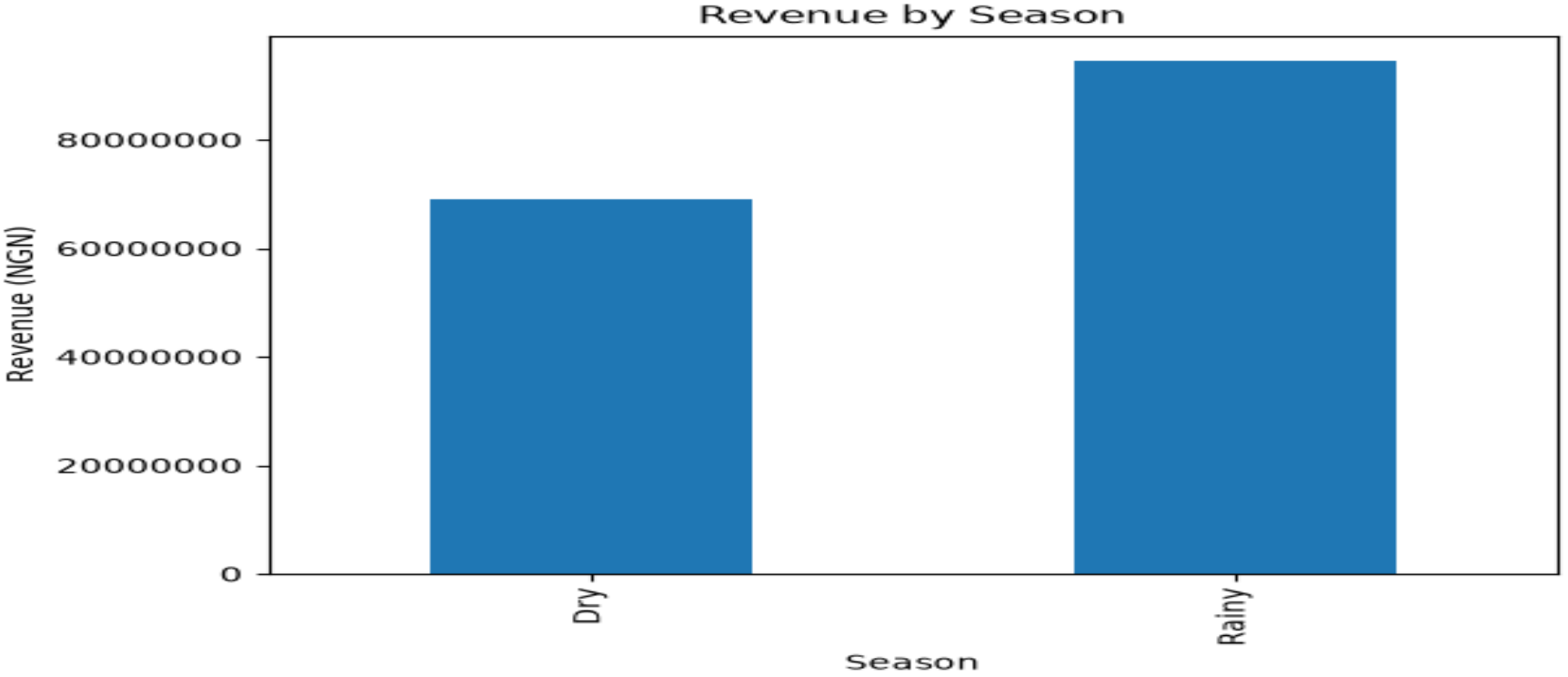
Monthly sales trend

- Monthly sales varied throughout the period. January 2024 recorded the highest revenue of ₦8.25 million, while June 2025 recorded the lowest at ₦4.89 million.



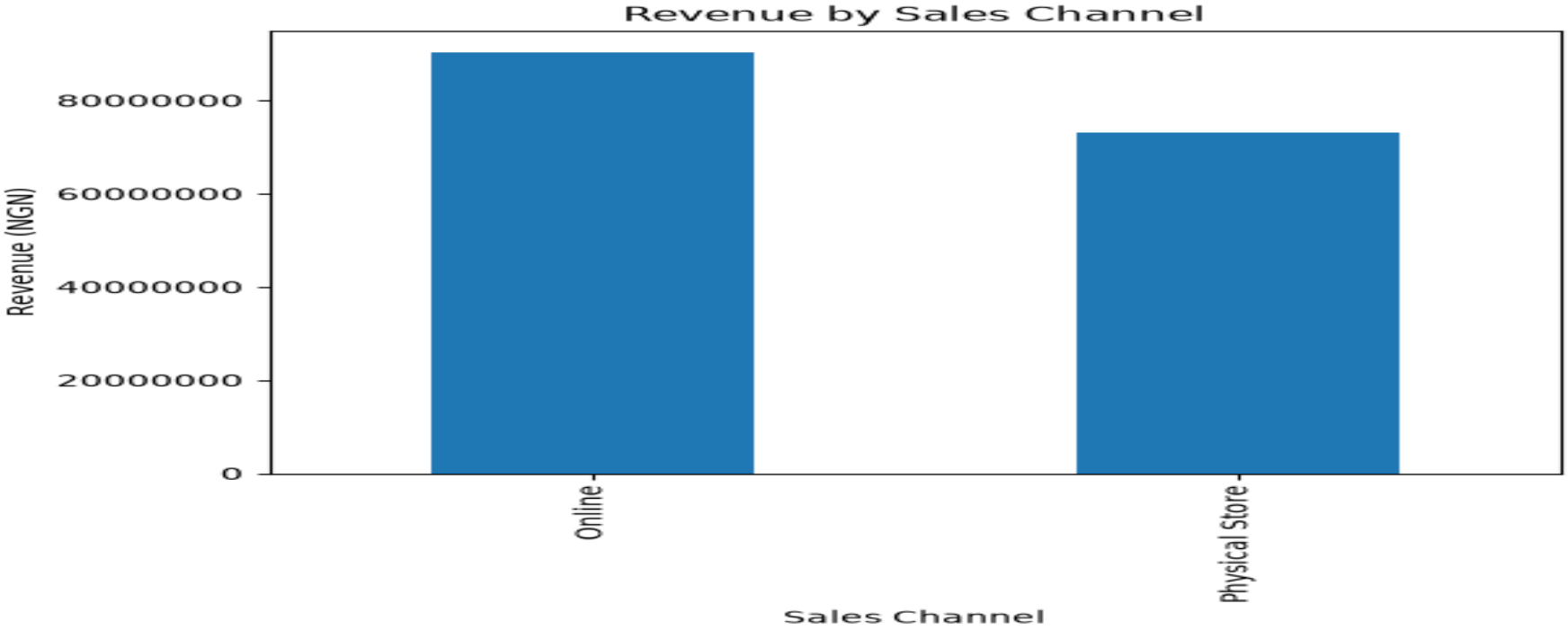
Seasonal sales performance

- The Rainy Season generated total revenue of ₦94.52 million, while the Dry Season generated ₦69.08 million. The company should plan stock and marketing based on seasonal demand.



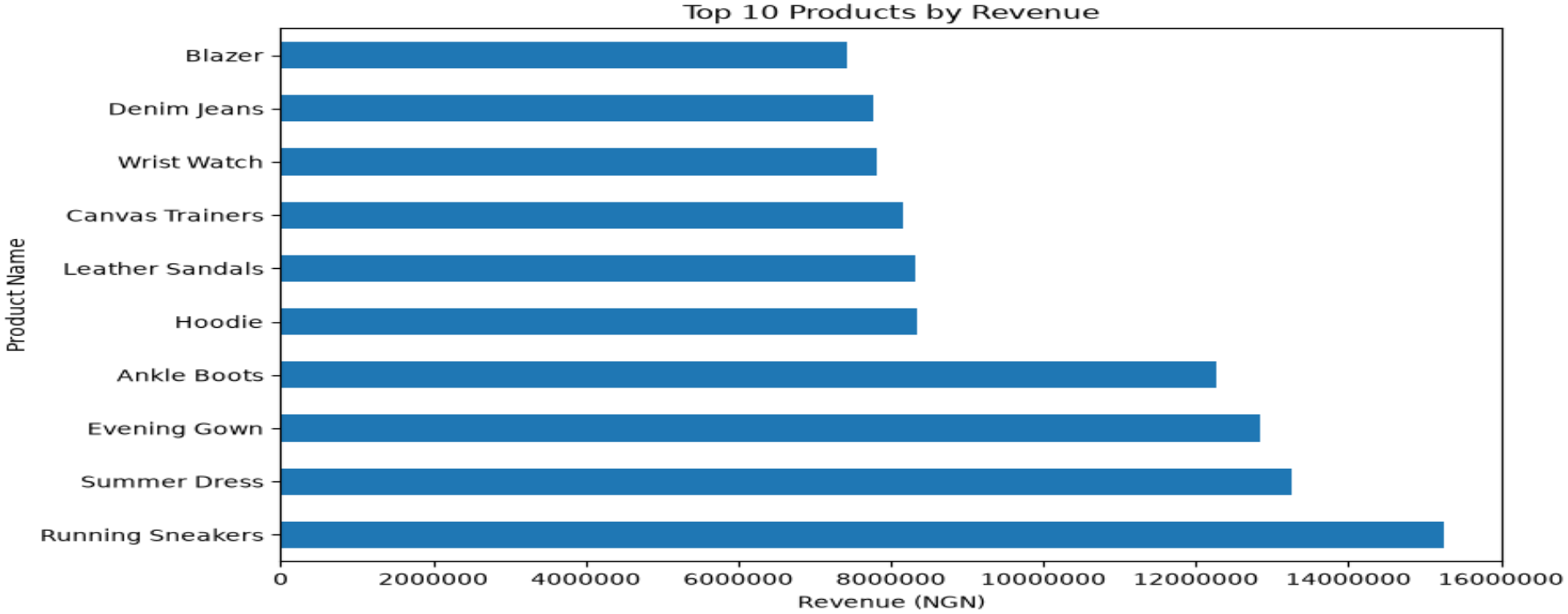
Online vs Physical Store Sales.

- Online sales generated ₦90.40 million, while physical store sales generated ₦73.21 million. This shows that online sales performed better. The company should continue improving its online shopping experience



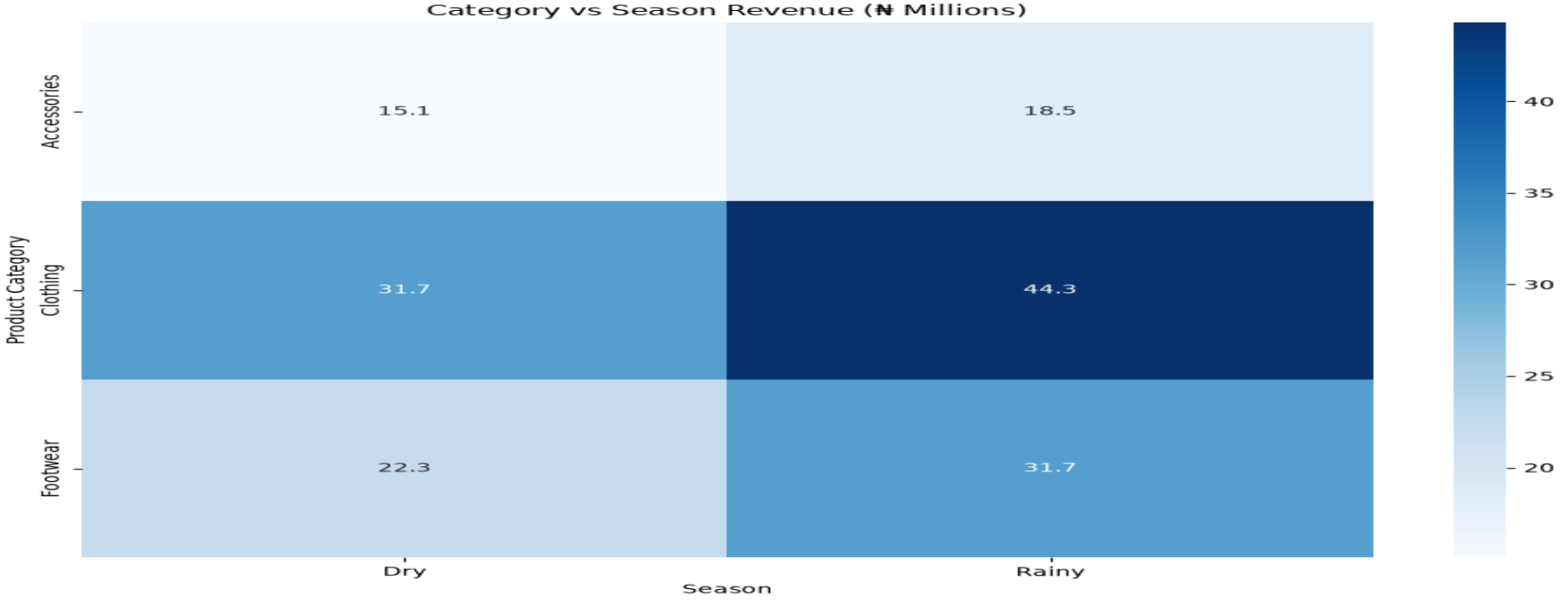
Top 10 Products by Revenue.

- Running Sneakers generated the highest revenue, followed by Summer Dress and Evening Gown. The company should maintain enough stock of popular products to meet customer demand.



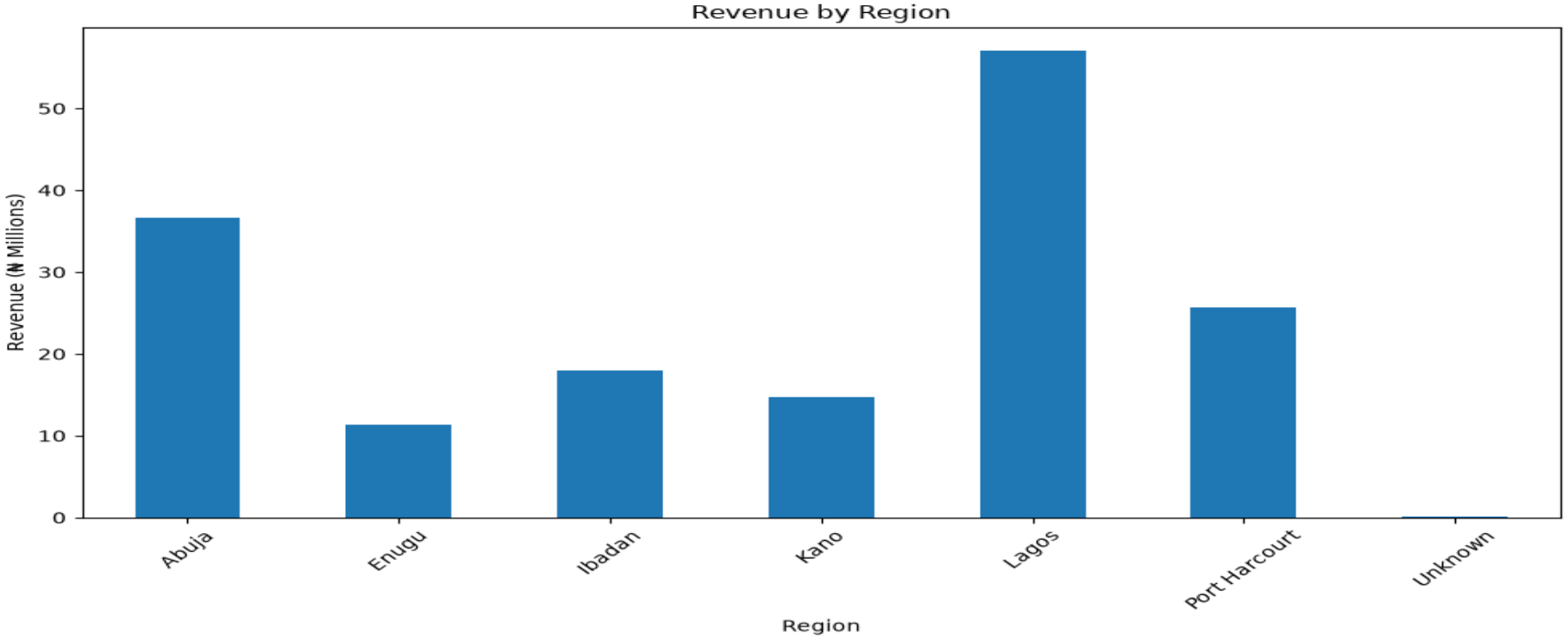
Category vs Seasonal Revenue.

- Clothing generated the highest revenue in both seasons. Footwear was the second-highest category, while Accessories generated the lowest revenue. Revenue was higher during the rainy season across all three categories.



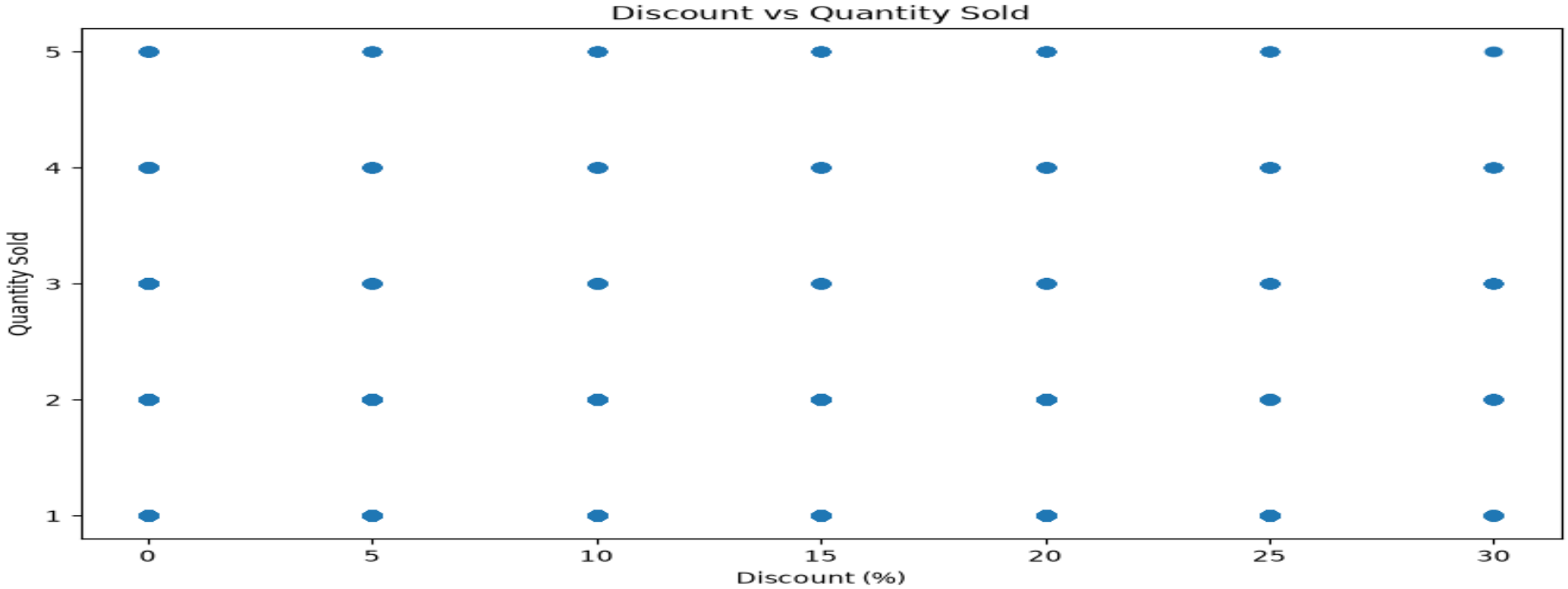
Regional Sales Performance.

- Sales performance varied across regions. The company should identify regions with lower sales and improve marketing efforts there while maintaining good service in strong-performing regions.



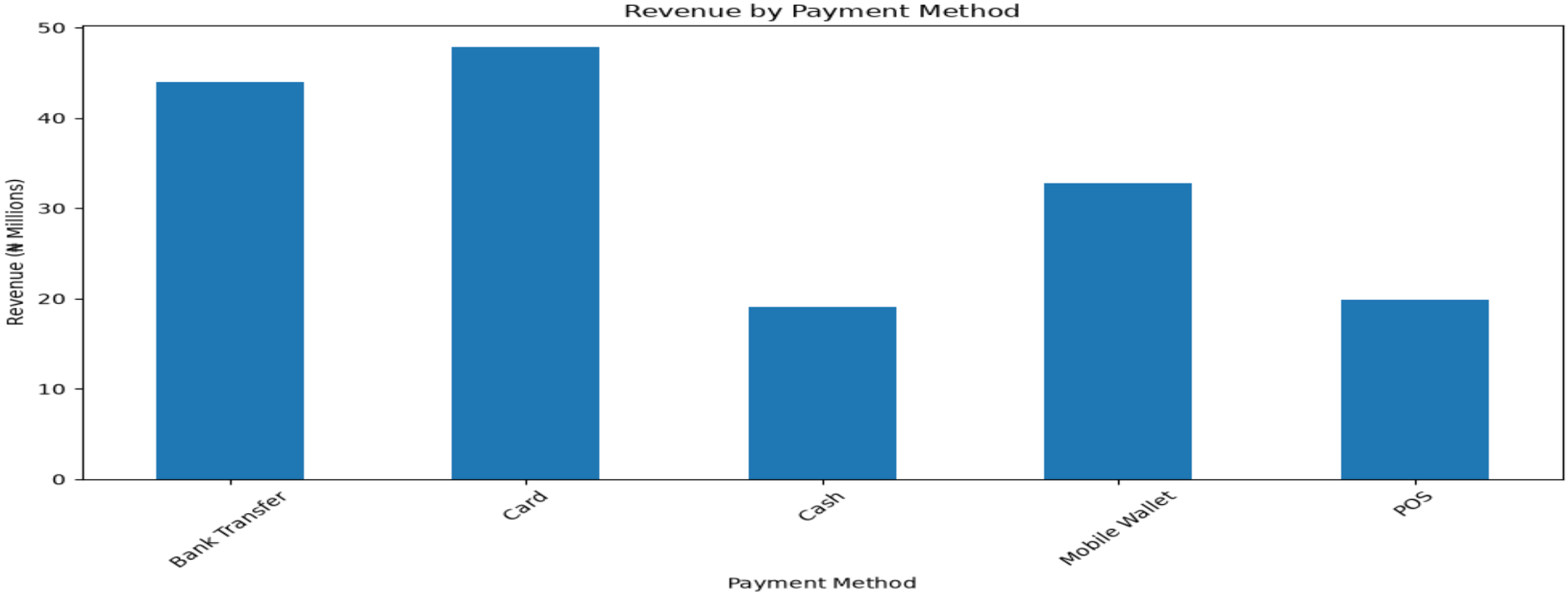
Discount vs Quantity Sold.

- The chart shows the relationship between discounts and the quantity of items sold. The company can use discounts carefully to encourage purchases while maintaining good revenue.



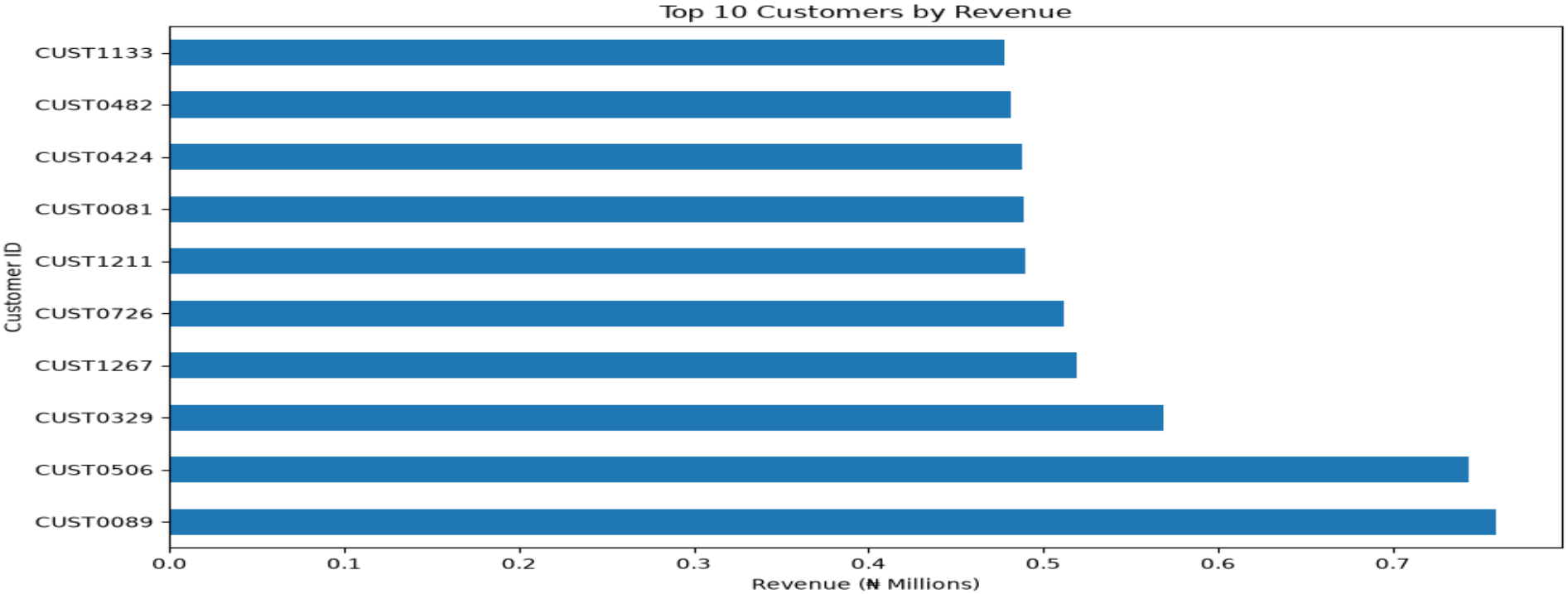
Revenue by Payment Method.

- The chart compares revenue generated through different payment methods. The company can use this information to understand customer payment preferences and make the payment process more convenient.



Top 10 Customers by Revenue.

- The chart identifies the customers who generated the highest revenue. The company can use this information to understand valuable customers and improve customer retention through good service and suitable promotions.



BUSINESS RECOMMENDATION

1. Stock more popular clothing products.
2. Use promotions to improve sales during slow months.
3. Plan inventory based on seasonal sales patterns.
4. Invest more in online marketing.
5. Maintain enough stock of top-performing products like Running Sneakers.

CONCLUSION

This project helped me understand fashion sales trends and customer purchasing behavior.

I found that Clothing generated the highest revenue, Online sales performed better than Physical Stores, and Running Sneakers were the top product.

The findings can help the company improve sales, marketing, and inventory planning.

Thank You!